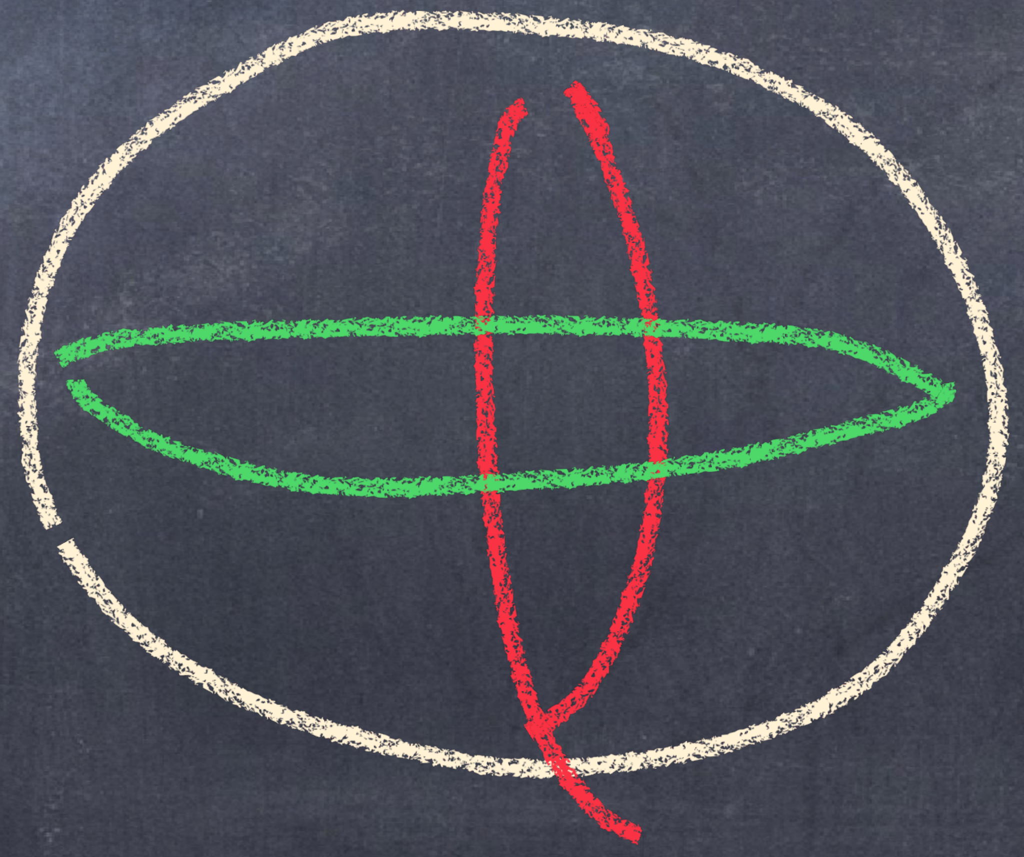
New question: can we recover SK_n^{∂} as π_0 of a K-theory spectrum?

Issue #2: assemblers & subtractive cats do not work

Problem:



covers do not have common subcovers!

Philosophy behind the problem

- classical K-theory (of exact, Waldhausen, ∞-stable cats) "$S_\bullet$-construction"
  encodes relations of the form $[A]+[C]=[B]$
  for exact seq $A \to B \to C$

- assemblers / subtractive cats frameworks
  encode relations of the form $[A]+[C]=[B]$
  for "subtractive seq" $A \to B \leftarrow C$

we really want to encode relations of the form
$[A]+[B]=[C]+[D]$ for

$$\begin{tikzcd} A \arrow[r] \arrow[d] & D \arrow[d] \\ B \arrow[r] & C \end{tikzcd}$$

K-theory of categories w/ squares

(Campbell + Zakharevich)

Def: A cat $\mathcal{C}$ w/ squares has

- a subcat of "cofibrations" $\rightarrowtail$
- a subcat of "cofiber maps" $\twoheadrightarrow$
- distinguished squares
$$\begin{tikzcd} A \arrow[r, tail] \arrow[d, tail] \arrow[dr, phantom, "\square"] & B \arrow[d, tail] \\ C \arrow[r, tail] & D \end{tikzcd}$$
s.t.

& 0 initial for $\rightarrowtail$

1) $\mathcal{C}$ has $\amalg$ and $\square$ closed under $\amalg$
2) $\square$ compose horizontally & vertically
3) $\rightarrowtail$, $\twoheadrightarrow$ contain all isos
4) if both $\parallel$ maps in a square are $\cong$, then the sq. is $\square$.

<u>Def</u>: $K^{\square}(\mathcal{C}) = \Omega_0 \left| p,q \mapsto \begin{array}{ccccccc} A_{11} & \rightarrowtail & A_{12} & \rightarrowtail & \cdots & \rightarrowtail & A_{1p} \\ \downarrow & \square & \downarrow & & & & \\ A_{21} & \rightarrowtail & \cdot & \cdot & \cdot & \cdot & \\ \vdots & & & & & & \\ \downarrow & \square & & & & \square & \downarrow \\ A_{q1} & \rightarrowtail & & \cdot & \cdot & \cdot & \cdot \end{array} \right|$

<u>Thm</u> (Campbell–Zakharevich)

$$K_0^{\square}(\mathcal{C}) = \frac{\text{free ab gp on ob of } \mathcal{C}}{[0]=0, \quad [A]+[D]=[B]+[C] \quad \text{for } \begin{array}{ccc} A & \rightarrowtail & B \\ \downarrow & \square & \downarrow \\ C & \rightarrowtail & D \end{array}}$$

& can see: this is an ∞ loop space

Def: $\mathrm{Mnfld}^{\partial}_{n}$ is a cat w/ squares w/

- $0 = \emptyset$
- horizontal = vertical maps
  = smooth emb. $M \hookrightarrow N$ s.t. ∂M maps to a submnfld of N w/ triv. normal bundle & each ∂ piece maps entirely to the interior or to a ∂ piece
- $$\begin{tikzcd} N \arrow[r, tail] \arrow[d, hook] & M \arrow[d, hook] \\ M' \arrow[r, tail] & M \cup_N M' \end{tikzcd}$$ are p.o.

<u>Thm</u>: (HMMRS)

$\mathrm{Mnfld}^\partial_n$ is a cat w/ squares &

$K_0^\square(\mathrm{Mnfld}^\partial_n) \cong SK^\partial_n$

<u>Thm</u>

$\exists$ map of spectra $K^\square(\mathrm{Mnfld}^\partial_n) \to K(\mathbb{Z})$

which on π_0 agrees w/ χ.

Proof idea:

Prop: $\mathcal{C}$ is Waldhausen cat

Then $\mathcal{C}^{\square}$ $\rightarrowtail$ = cof

$\twoheadrightarrow$ = all maps

$$\begin{array}{ccc} A & \rightarrowtail & B \\ \downarrow & & \downarrow \\ C & \rightarrowtail & D \end{array} = \text{sq. w/ } C \cup_A B \xrightarrow{\simeq} D$$

is a cat w/ squares & $K^{\square}(\mathcal{C}) \simeq K^{\text{Wald}}(\mathcal{C})$

(Thomason). Γ & reduce to Mayer-Vietoris